2026학년도 1학기 수학1 Test2 시험지/답안지

시행일시: 2026년 6월 1일(월) (70분간)

과목명	수강번호	학부/학과: 학년:	이름	(이름은 반듯하게 크게 쓰세요.)	감독자인	점수
수학1	X____	학번:				

I.(40pts) Fill in the blanks. (단답형: 답을 반드시 빈칸에 적으십시오.)

1.(2+2pts) (1) Find all points whose polar coordinates lies in the second quadrant. ③④

① $\left(-1, \frac{\pi}{4}\right)$ ② $\left(1, -\frac{3\pi}{4}\right)$ ③ $\left(-1, \frac{7\pi}{4}\right)$ ④ $\left(1, \frac{11\pi}{4}\right)$

(2) The polar equation $r = -4\sin\theta + 8\cos\theta$ represents a circle. Find its center and radius R. Center : (4 , -2) and $R =$ $2\sqrt{5}$

2.(2+3pts) (1) If the nth partial sum of a series $\sum_{n=1}^{\infty} a_n$ is $S_n = \frac{n-1}{4n+1}$, then $\sum_{n=1}^{\infty} a_n =$ $\frac{1}{4}$

(2) $\sum_{n=1}^{\infty}\left(\tan^{-1}(n+1) - \tan^{-1}(n)\right) =$ $\frac{\pi}{4}$

3.(2pts) Suppose f is a continuous positive decreasing function for $x \geq 1$ and $a_n = f(n)$. Write the following three quantities in increasing order: $\int_1^{2026} f(x)\,dx$, $\sum_{n=1}^{2025} a_n$, $\sum_{n=2}^{2026} a_n$

$\sum_{n=2}^{2026} a_n$ $\leq$ $\int_1^{2026} f(x)\,dx$ $\leq$ $\sum_{n=1}^{2025} a_n$

4.(5pts) Choose all correct statements. ①②④

① If $\{a_n\}$ is an increasing sequence and all its terms lie between 2025 and 2026, then the sequence is convergent.

② If $a_n \geq 0$ and $\sum a_n$ converges, then $\sum a_n^2$ also converges.

③ If $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ is convergent.

④ If a power series $\sum_{n=0}^{\infty} c_n (x-9)^n$ converges at $x = 6$, then $\sum_{n=0}^{\infty} c_n$ converges.

⑤ If the radius of convergence of a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ is 2, then $\sum_{n=0}^{\infty} \frac{2n}{n+1} c_n (x-a)^n$ has the radius of convergence 1.

5.(5pts) Choose all convergent series. ①②③④

① $\sum_{n=1}^{\infty} \frac{n^2+1}{n^2}\left(\frac{1}{2}\right)^n$ ② $\sum_{n=1}^{\infty} \frac{\sqrt[3]{n}}{\sqrt{n^3+2n+1}}$ ③ $\sum_{n=1}^{\infty} \frac{9^n}{3+10^n}$

④ $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ ⑤ $\sum_{n=2}^{\infty} \frac{1}{n}\left(1-\frac{1}{n}\right)^n$

6.(4pts) Determine whether the series is absolutely convergent, conditionally convergent, or divergent. Circle your answer.

(1) $\sum_{n=2}^{\infty} (-1)^n \frac{1}{\sqrt{n}\ln n}$ (abs. converges, cond. converges, diverges)

(2) $\sum_{n=2}^{\infty} (-1)^n \ln\left(\frac{n^2}{2n^2-5}\right)$ (abs. converges, cond. converges, diverges)

(3) $\sum_{n=1}^{\infty} \frac{\sin n}{n^{3/2}}$ (abs. converges, cond. converges, diverges)

(4) $\sum_{n=1}^{\infty} (-1)^n \sin\left(\frac{1}{3^n}\right)$ (abs. converges, cond. converges, diverges)

7.(2+2pts) Find all values of p for which each of the following series is convergent.

(1) $\sum_{n=2}^{\infty} \frac{(\ln n)^p}{n}$ $p < -1$ (2) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ $p > 0$

8.(2+2pts) (1) If $e^x = \sum_{n=0}^{\infty} c_n (x-2)^n$, then $c_7 =$ $\frac{e^2}{7!}$

(2) If $e^x + e^{-x} = \sum_{n=0}^{\infty} d_n (x-2)^n$, then $d_7 =$ $\frac{e^2 - e^{-2}}{7!}$

9.(4+3pts) Suppose that $\frac{x}{5+x^3} = \sum_{n=0}^{\infty} a_n x^n$ for $|x| < R$, where R is the radius of convergence of the power series.

(1) $a_{10} =$ $-\frac{1}{5^4}$ and $R =$ $\sqrt[3]{5}$ or $5^{\frac{1}{3}}$

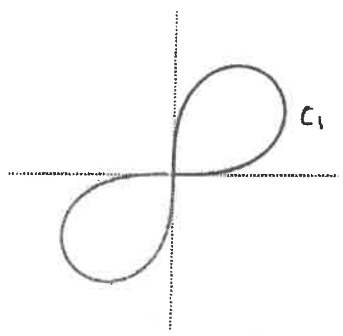
(2) $\int_0^{0.1} \frac{x}{5+x^3}\,dx \approx$ $\frac{1}{5\cdot 2}\left(\frac{1}{10}\right)^2 - \frac{1}{5^2}\left(\frac{1}{10}\right)^5$ $-\frac{1}{625}$ (|error| < 10^{-10})

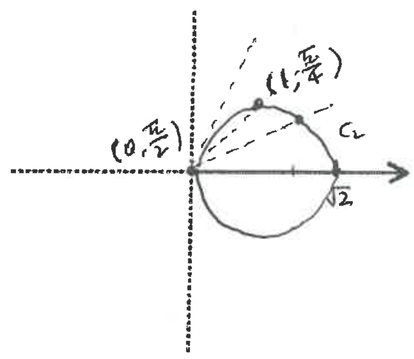
※문제 II ~ VI은 서술형입니다. 답안지에 풀이와 함께 답을 적으십시오.

II.(2+4pts) Consider the polar curves

$$C_1 : r^2 = \sin(2\theta) \quad \text{and} \quad C_2 : r = \sqrt{2}\cos\theta.$$

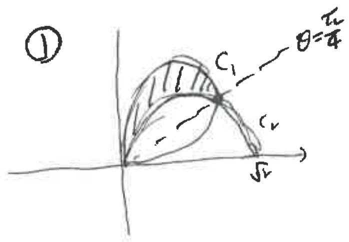
1. The graph of C_1 is shown below. Sketch C_2 on the axes to the right.





2. Find the area of the region that lies inside the curves C_1 and outside the curve C_2.

①



$r^2 = \sin(2\theta)$, $r = \sqrt{2}\cos\theta$ 의 교점

$\sin(2\theta) = 2\cos^2\theta$

$2\sin\theta\cos\theta = 2\cos^2\theta$

$\cos\theta(\sin\theta - \cos\theta) = 0$

$\theta = \frac{\pi}{2}$, $\theta = \frac{\pi}{4}$

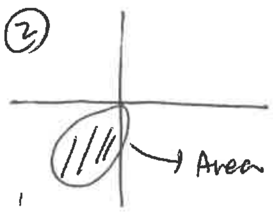
area $A = \frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin(2\theta) - 2\cos^2\theta)\,d\theta$ $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$= \frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin 2\theta - 1 - \cos 2\theta)\,d\theta$

$= \frac{1}{2}\left[-\frac{1}{2}\cos 2\theta - \theta - \frac{1}{2}\sin 2\theta\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

$= \frac{1}{2}\left[\left(\frac{1}{2} - \frac{\pi}{2}\right) - \left(-\frac{\pi}{4} - \frac{1}{2}\right)\right] = \frac{1}{2}\left(1 - \frac{\pi}{4}\right) = \frac{1}{2} - \frac{\pi}{8}$ ///

②



Area $= \frac{1}{2}\int_{\pi}^{\frac{3}{2}\pi} \sin(2\theta)\,d\theta$

$= \frac{1}{2}\left[-\frac{1}{2}\cos(2\theta)\right]_{\pi}^{\frac{3}{2}\pi} = \frac{1}{2}$

③ $1 - \frac{\pi}{8}$ ///

<뒷면계속>

Ⅲ.(5pts) Determine whether the improper integral $\int_0^\infty e^{-x^2}\,dx$ is convergent or divergent.

$$\int_0^\infty e^{-x^2}dx = \underbrace{\int_0^1 e^{-x^2}dx}_{\text{정적분. 수렴}} + \int_1^\infty e^{-x^2}dx$$

$x \geq 1$ 이면 $x^2 \geq x$

$-x^2 \leq -x$

$0 < e^{-x^2} \leq e^{-x}$

$$\int_1^\infty e^{-x}dx = \lim_{b\to\infty}\int_1^b e^{-x}dx = \lim_{b\to\infty}\left[-e^{-x}\right]_1^b$$

$$= \lim_{b\to\infty} e^{-1} - e^{-b} = e^{-1} \text{ converges.}$$

By the Comparison Test, $\int_1^\infty e^{-x^2}dx$ converges

∴ $\int_0^\infty e^{-x^2}dx$ is convergent ///

Ⅳ.(4+4pts) Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

1. $\sum_{n=1}^{\infty} \frac{(-1)^n(1+e^{1/n})}{\sqrt{n}}$

① $\frac{1+e^{1/n}}{\sqrt{n}} > \frac{1}{\sqrt{n}}$ & $\sum_{n=1}^\infty \frac{1}{\sqrt{n}}$ is divergent p-series with $p=\frac{1}{2}\leq 1$.

By DCT, $\sum \frac{1+e^{1/n}}{\sqrt{n}}$ is divergent.

② $b_n = \frac{1+e^{1/n}}{\sqrt{n}} \geq b_{n+1} = \frac{1+e^{\frac{1}{n+1}}}{\sqrt{n+1}}$ for all n

& $\lim_{n\to\infty} b_n = \lim_{n\to\infty}\frac{1+e^{1/n}}{\sqrt{n}} = 0$. By AST, $\sum \frac{(-1)^n(1+e^{1/n})}{\sqrt{n}}$ converges

∴ conditionally convergent ///

2. $\sum_{n=1}^{\infty} \frac{\cos 2n}{1+2^n}$

$\frac{|\cos 2n|}{1+2^n} \leq \frac{1}{2^n}$ for all n

& $\sum_{n=1}^\infty \left(\frac{1}{2}\right)^n$ is a convergent geometric series with $r=\frac{1}{2}$.

By DCT, $\sum \frac{\cos 2n}{1+2^n}$ is absolutely convergent ///

Ⅴ.(5pts) Show that the series is convergent. How many terms of the series do we need to add in order to find the sum to the indicated accuracy?

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}n}{2^n} \quad (|\text{error}| < 0.07)$$

① $b_n = \frac{n}{2^n} \geq b_{n+1} = \frac{n+1}{2^{n+1}}$ for all n. $\left(\because \frac{b_{n+1}}{b_n} = \frac{(n+1)}{2^{n+1}}\cdot\frac{2^n}{n} = \frac{n+1}{2n} \leq 1\right)$

② $\lim_{n\to\infty} b_n = \lim_{n\to\infty}\frac{n}{2^n} = 0$ $\left(\lim_{x\to\infty}\frac{x}{2^x} = 0 \text{ 로피탈}\right)$

By AST, the given series is convergent.

② $\sum_{n=1}^\infty \frac{(-1)^{n+1}n}{2^n} = \frac{1}{2} - \frac{2}{2^2} + \frac{3}{2^3} - \frac{4}{2^4} + \frac{5}{2^5} - \frac{6}{2^6} + \frac{7}{2^7} \cdots$

$\frac{6}{2^6} = 6\cdot\frac{1}{64}$, $\frac{7}{2^7} = 7\cdot\frac{1}{128} \leq 7\cdot\frac{1}{100} = 0.07$ error

∴ $N=6$

6 terms. We need to add 6 terms.

Ⅵ.(6pts) Find the radius of convergence and interval of convergence of the power series $\sum_{n=0}^{\infty}\frac{(x+1)^n}{3^{n+1}\sqrt{n+1}}$.

$a_n = \frac{(x+1)^n}{3^{n+1}\sqrt{n+1}}$

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(x+1)^{n+1}}{3^{n+2}\sqrt{n+2}}\cdot\frac{3^{n+1}\sqrt{n+1}}{(x+1)^n}\right| = \frac{1}{3}|x+1| \begin{cases} <1 & \text{conv} \\ >1 & \text{div} \end{cases}$ by the Ratio test

∴ $|x+1| < 3$, radius of convergence $R=3$ ///

$-3 < x+1 < 3$

$-4 < x < 2$

$x=2$ 일 때 $\sum_{n=0}^\infty \frac{3^n}{3^{n+1}\sqrt{n+1}} = \sum_{n=0}^\infty \frac{1}{3\sqrt{n+1}}$ $= \frac{1}{3}\sum_{n=1}^\infty\frac{1}{\sqrt{n}}$ diverges ∵ $p=\frac{1}{2}<1$, p-series

$x=-4$ 일 때 $\sum_{n=0}^\infty\frac{(-3)^n}{3^{n+1}\sqrt{n+1}} = \sum_{n=0}^\infty \frac{(-1)^n}{3\sqrt{n+1}}$ converges by AST

$\left(\begin{array}{l} ① \ b_n = \frac{1}{\sqrt{n+1}} \geq b_{n+1} = \frac{1}{\sqrt{n+2}} \\ ② \ \lim_{n\to\infty} b_n = 0 \end{array}\right)$

∴ interval of convergence $[-4, 2)$ ///

묶는 곳입니다. 답을 쓰지 마십시오.

1) 서술형문제는 답과 함께 풀이도 답안지에 적었습니까?

2) 연습장에 쓴 것을 옮겨 적었습니까? 수고하셨습니다.